

# Parth Samir Dalvi

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## PROFESSIONAL EXPERIENCE

### Application Developer

May 2025 – May 2026

*Gameplay Inc*

*California, USA*

- Engineered scalable AI features, modular UI components, and REST API services using C#/.NET and Python, reducing client-side rendering latency by **18%** across high-traffic enterprise endpoints.
- Optimized complex relational database queries, indexing strategies, and ETL pipelines in PostgreSQL, achieving a **20% performance boost** across core analytical workloads and background jobs.
- Led PyTest and Jest automated testing suites alongside peer code reviews within a cross-functional team, catching 15+ critical edge cases and boosting code coverage by **12%** prior to release.

### Data Analyst

March 2022 – July 2024

*Weagile Software Solutions*

*Maharashtra, India*

- Designed end-to-end data visualization pipelines and interactive Tableau/Power BI dashboards, processing high-volume behavioral metrics to deliver real-time operational insights to leadership.
- Built LLM-driven automation scripts using chain-of-thought prompting to accelerate analytical pipeline velocity, streamline data transformation steps, and automate recurring dashboard updates.
- Streamlined asynchronous ETL data workflows, cutting manual reporting cycles by **40%** while enforcing strict data validation against upstream production SQL databases and API sources.

## TECHNICAL SKILLS

**Languages:** Python, C++, TypeScript, JavaScript, C#, SQL, HTML5/CSS3, Shell/Bash

**AI/ML & Edge AI:** PyTorch, ONNX Runtime, Model Quantization (INT4/INT8), YOLOv8, OpenCV, RAG, NLP, Edge AI

**Agentic AI & LLMops:** Multi-Agent Systems, LangChain, LangGraph, Vector DBs (SQLite-VSS, pgvector, Pinecone), GPT-4, Gemini API

**Frameworks & Web:** Next.js, Node.js, Spring Boot, FastAPI, Flask, React, React Native, Angular Signals, Electron, WebSockets

**Systems, Data & MLOps:** Algorithms (A\* Search, Dijkstra's), MLOps, Time-Series Forecasting, ETL Pipelines, PortAudio, Azure, Docker

## EDUCATION

### Purdue University

Indiana, USA

*Master of Science in Computer Science*

August 2024 – May 2026

Software Engineering, Distributed Systems, Cloud Computing, Advanced Algorithms, Machine Learning, Database Systems

### University of Pune

Maharashtra, India

*Bachelor of Technology in Computer Science*

August 2020 – July 2024

Data Structures and Algorithms, Operating Systems, Database Management Systems, Object Oriented Programming, Computer Networks

## KEY PROJECTS

### Fibucks | [fibucks.com](https://fibucks.com)

Aug. 2025 – Feb. 2026

*Next.js, TypeScript, MongoDB, Node.js, Spring Boot, Azure, PyTorch, RAG, Vector DBs*

- Built and deployed a full-stack personal finance platform with Node.js and Spring Boot microservices on Azure.
- Secured user sessions with OAuth 2.0/JWT authentication, achieving 100% data isolation and OWASP compliance.
- Shipped an AI advisor feature using RAG and pgvector semantic search to surface predictive spending alerts.
- Engineered analytical data pipelines and Chart.js reporting interfaces to visualize user cash flow metrics.

### Autonomous Drone Swarm Systems

Jan. 2024 – Dec. 2024

*Independent Project & IAF-Partnered Initiative | Python, ROS, OpenCV, C++, Edge AI, Distributed Systems*

- Built decentralized swarm coordination logic across 50+ units for mesh sync and collision avoidance.
- Deployed real-time computer vision models on Raspberry Pi hardware for localization and threat detection.
- Co-led a defense extension with the Indian Air Force, ranking in the **top 1%** and earning an IAF Appreciation Letter.
- Named National Grand Finalist in a defense hackathon for autonomous flight logic and fault-tolerant design.

### Real-Time Audio-Enabled Object Detection System

Jan. 2024 – Dec. 2024

*Python, YOLOv8, NLP, Gemini API, PyTorch, Model Evaluation, Systems Testing*

- Engineered a real-time computer vision pipeline with YOLOv8, reaching 98% mAP accuracy in field tests.
- Integrated voice NLP via Gemini API to stream spoken object feedback for assistive-technology applications.
- Co-authored a peer-reviewed research paper on edge inference optimization, published in **IJSREM** journal.
- Constructed evaluation benchmark suites to measure inference speed, latency, and model precision under load.

## HONORS & LEADERSHIP

**National Grand Finalist, Smart India Hackathon:** Selected out of thousands of applicants nationwide for top-tier execution in physical hardware system prototyping and autonomous algorithm delivery (Hardware Edition).

**Formal Appreciation Letter, Indian Air Force:** Recognized by an IAF Squadron Leader for engineering execution on a mission-critical autonomous drone initiative.

**Technical Team Volunteer Lead, Helping Hearts NGO:** Managed infrastructure logistics, hardware deployments, and localized network setup support for community outreach.